Module 13: Dashboard Design & User Experience (UX)

📖 Overview

Creating charts is easy. Creating dashboards that executives and stakeholders actually use is much harder.

A successful dashboard is not measured by the number of visuals it contains. It is measured by how effectively it answers business questions, communicates insights, and supports decision-making.

This module focuses on dashboard design principles, user experience (UX), information architecture, KPI placement, navigation, and professional dashboard development techniques used in enterprise environments.

🎯 Learning Objectives

After completing this module, you will be able to:

- Understand dashboard design principles
- Plan dashboards before development
- Design executive-level dashboards
- Apply UX principles in Power BI
- Create effective KPI layouts
- Build intuitive navigation systems
- Design mobile-friendly dashboards
- Improve dashboard usability and performance

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1️⃣ Introduction to Dashboard Design

What is a Dashboard?

A dashboard is a single-screen visual interface that presents the most important information needed to achieve one or more objectives.

Goal of a Dashboard

A dashboard should:

- Monitor performance
- Identify issues
- Track KPIs
- Support decisions

Dashboard Philosophy

A user should understand the health of a business within a few seconds of opening the dashboard.

2️⃣ What Makes a Good Dashboard?

A good dashboard answers business questions quickly.

Characteristics

Clear

Easy to understand.

Relevant

Contains only necessary information.

Interactive

Supports exploration.

Actionable

Leads to business decisions.

Consistent

Uses standardized design elements.

3️⃣ Dashboard vs Report

Many beginners confuse dashboards and reports.

Dashboard

High-level overview.

Focuses on KPIs.

Example

Executive Sales Dashboard.

Contains:

- Revenue
- Profit
- Growth %
- Top Products

Report

Detailed analysis.

Contains:

- Tables
- Drill-through pages
- Detailed transactions

Comparison

Feature	Dashboard	Report
Purpose	Monitoring	Analysis
Detail Level	Summary	Detailed
Pages	Usually One	Multiple
Audience	Executives	Analysts
Interaction	Moderate	High

4️⃣ Understanding Your Audience

Dashboard design begins with understanding users.

Executive Users

Need:

- KPIs
- Trends
- High-level insights

Managers

Need:

- Department Performance
- Team Metrics
- Operational Insights

Analysts

Need:

- Detailed Data
- Drill-through Reports
- Advanced Filtering

Key Principle

Different audiences require different dashboards.

5️⃣ Dashboard Planning Process

Never start by dragging visuals onto a blank canvas.

Step 1: Define Objectives

Example:

```
Monitor Sales Performance
```

Step 2: Identify KPIs

Examples:

- Revenue
- Profit
- Growth %
- Customer Count

Step 3: Define Questions

Example:

```
Which products drive revenue?
```

Step 4: Sketch Layout

Create a wireframe before building.

Step 5: Build Dashboard

Implement visuals.

6️⃣ Information Architecture

What is Information Architecture?

Information Architecture determines how information is organized and presented.

Common Dashboard Flow

```
KPIs
   ↓
Trends
   ↓
Category Analysis
   ↓
Detailed Data
```

User Reading Pattern

Most users follow:

F-Pattern

or

Z-Pattern

when scanning dashboards.

7️⃣ Visual Hierarchy

Visual Hierarchy guides user attention.

Most Important Information

Should appear:

```
Top Left
Top Center
```

Typical Layout

```
KPIs
 ↓
Trend Analysis
 ↓
Detailed Breakdown
 ↓
Tables
```

Why Important?

Users naturally focus on large and prominent elements first.

8️⃣ KPI Design

What is a KPI?

KPI = Key Performance Indicator

Measures business success.

Example KPIs

- Revenue
- Profit
- Profit Margin
- Customer Retention
- Attrition Rate

Effective KPI Card

Should contain:

Current Value

```
₹5.2M
```

Trend

```
↑ 12%
```

Target

```
Target: ₹5.0M
```

Status Indicator

```
Green = Good
Red = Risk
```

9️⃣ Layout Design Principles

Rule 1

Keep dashboard on a single screen whenever possible.

Rule 2

Group related visuals together.

Rule 3

Use consistent spacing.

Rule 4

Maintain alignment.

Rule 5

Use white space strategically.

Example Layout

```
----------------------------------
 KPI  KPI  KPI  KPI
----------------------------------
 Revenue Trend
----------------------------------
 Product Analysis | Region Analysis
----------------------------------
 Detailed Table
----------------------------------
```

🔟 Color Strategy

Colors should support communication.

Primary Uses

Highlight Important Information

Example:

```
Revenue Growth
```

Show Status

Example:

```
Green = Positive
Red = Negative
```

Differentiate Categories

Example:

```
Electronics
Furniture
Clothing
```

Best Practices

Limit Colors

Use 4–6 primary colors.

Stay Consistent

Same categories should always use the same colors.

Avoid Rainbow Dashboards

Too many colors create confusion.

1️⃣1️⃣ User Experience (UX)

What is UX?

UX = User Experience

Focuses on how users interact with dashboards.

Good UX Characteristics

Easy Navigation

Users find information quickly.

Consistent Design

Predictable interactions.

Fast Performance

Visuals load quickly.

Minimal Cognitive Load

Users don't have to think excessively.

UX Goal

Reduce effort required to find insights.

1️⃣2️⃣ Navigation Design

Large reports require navigation systems.

Navigation Methods

Buttons

Example:

```
Home
Sales
```

```
Finance
HR
```

Bookmarks

Switch between dashboard views.

Page Navigation

Move between report pages.

Drill-through

Navigate to detailed reports.

1️⃣3️⃣ Interactive Dashboard Design

Interactivity improves exploration.

Slicers

Allow filtering.

Examples:

- Date
- Product
- Region

Drill Down

Move through hierarchies.

Example:

```
Year
 ↓
Quarter
 ↓
Month
```

Tooltips

Provide additional information.

Cross Filtering

Visuals interact automatically.

1️⃣4️⃣ Mobile Dashboard Design

Many users consume reports on phones.

Mobile Design Principles

Simplify Layout

Avoid excessive visuals.

Use Vertical Structure

Stack visuals.

Prioritize KPIs

Show most important metrics first.

Use Larger Fonts

Improve readability.

Power BI Mobile Layout

Provides dedicated mobile view.

1️⃣5️⃣ Executive Dashboard Design

Executives need quick insights.

Typical Executive Dashboard

KPI Section

- Revenue

- Profit
- Growth %

Trend Section

- Monthly Revenue

Business Drivers

- Top Products
- Top Regions

Risk Indicators

- Declining Sales
- Margin Reduction

Executive Dashboard Rule

Less detail.

More insights.

1️⃣6️⃣ Dashboard Performance Considerations

Poor design affects performance.

Common Causes

Too Many Visuals

Each visual creates queries.

Excessive Slicers

Increases interactions.

Complex DAX

Slows rendering.

Large Datasets

Increase load times.

Performance Guidelines

Limit visuals per page.

Typical recommendation:

```
6–12 visuals
```

1️⃣7️⃣ Common Dashboard Mistakes

Too Many Charts

Creates clutter.

No Business Focus

Displays data without purpose.

Poor Color Choices

Confuses users.

Missing KPI Context

No targets or comparisons.

Inconsistent Layout

Reduces usability.

Overusing Pie Charts

Makes comparison difficult.

Excessive Scrolling

Reduces user engagement.

1️⃣8️⃣ Best Practices

Start with Business Questions

Design around decisions.

Prioritize KPIs

Place them at the top.

Use Consistent Design

Fonts, colors, spacing.

Minimize Visual Clutter

Remove unnecessary elements.

Optimize Performance

Keep reports responsive.

Test with Users

Gather feedback before deployment.

📌 Key Terminologies

Term	Definition
Dashboard	High-level business overview
KPI	Key Performance Indicator
UX	User Experience
Information Architecture	Organization of information
Visual Hierarchy	Arrangement by importance
Navigation	User movement through reports
Drill-through	Detailed report navigation
Bookmark	Saved report state
Wireframe	Dashboard blueprint

Term	Definition
Cognitive Load	Mental effort required

📝 Practice Exercises

Exercise 1

Design a wireframe for a Sales Dashboard.

Exercise 2

Identify KPIs for:

- HR Dashboard
- Finance Dashboard
- Marketing Dashboard

Exercise 3

Evaluate an existing dashboard and identify UX issues.

Exercise 4

Create a navigation structure for a 5-page report.

Exercise 5

Design a mobile-friendly dashboard layout.

🎯 Mini Project

Executive Sales Dashboard Design

Objective

Design a dashboard before building it.

Requirements

KPI Section

- Revenue

- Profit
- Profit Margin
- Customer Count

Trend Section

- Monthly Revenue Trend

Analysis Section

- Product Performance
- Regional Performance

Filters

- Year
- Region
- Product Category

Deliverables

1. Dashboard Wireframe
2. KPI Placement Plan
3. Color Strategy
4. Navigation Plan
5. Mobile Layout Design

📚 Module Summary

In this module, you learned:

✅ Dashboard Design Principles

✅ Dashboard vs Report

✅ Audience Analysis

✅ Dashboard Planning

✅ Information Architecture

✅ Visual Hierarchy

✅ KPI Design

✅ Layout Design

✅ Color Strategy

✅ User Experience (UX)

✅ Navigation Design

✅ Mobile Dashboard Design

✅ Executive Dashboard Design

✅ Dashboard Performance Optimization

A well-designed dashboard communicates insights faster, improves decision-making, and significantly increases user adoption. These design principles are used by professional Power BI developers and BI teams in enterprise environments.

⏭️ Next Module

Module 14: Advanced Visualizations in Power BI

Topics Covered:

- Waterfall Charts
- Funnel Charts
- Scatter Charts
- Treemaps
- Gauges
- Ribbon Charts
- Decomposition Tree
- Key Influencers
- AI Visuals
- Custom Visuals
- Advanced Analytical Reporting